Mohamed Amin Ellithy

Data Analyst | SQL | Python | Power BI | Tableau | ETL & Reporting Automation
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PROFESSIONAL SUMMARY

Data Analyst with 3+ years in data analysis, reporting automation and machine learning, including 2+ years at Nokia turning large-scale telecom datasets into automated KPI reports and dashboards. Built Python and SQL ETL pipelines and data quality checks that cut manual reporting workload by 85% and reduced a 6-hour processing cycle to 20 minutes. Skilled in SQL, Python (pandas), Power BI (DAX, Power Query), Tableau, statistics and forecasting, turning business questions into actionable insights for stakeholders. Relocated to Al Khobar in 2026; MSc Data Science candidate with a transferable Iqama, available immediately.

SKILLS

Data Analysis: SQL (querying and data transformation), Python (pandas, NumPy), Exploratory Data Analysis (EDA), Statistical Analysis, Hypothesis Testing, Regression, Causal Inference (fixed effects), Root-Cause Analysis, Time Series Forecasting (SARIMA, Prophet), Predictive Analytics
Business Intelligence & Data Visualization: Power BI (DAX, Power Query), Tableau, Plotly, Matplotlib, Seaborn, Dashboard Design, KPI Reporting, Data Visualization, Microsoft Excel
Data Engineering & Data Quality: ETL Pipelines, Data Warehousing (SQL Server, Star Schema), Data Modeling, Data Preparation, Data Cleansing, Data Validation, Data Quality Checks, Relational Databases (SQL Server, PostgreSQL)
Automation & Tools: Reporting Automation, Process Automation, Web Scraping, Git, GitHub, Jupyter, Flask, Microsoft Azure, AI-Assisted Analytics (LLM tools)
Ways of Working: Stakeholder Management, Requirements Gathering, Communication of Findings, Technical Documentation, Training & Mentoring, Agile Collaboration

WORK EXPERIENCE

Automation and Analytics Engineer
Nokia - Cairo, Egypt | Sep 2023 - Jan 2026

- Designed and implemented an automated KPI reporting pipeline (ETL) in Python and SQL, integrating NetAct network data across 5G, 4G, 3G and 2G and cutting manual reporting workload by 85%.
- Built a self-service reporting engine in Python and Plotly that converts raw Excel templates into validated reports with 10+ chart types, giving non-technical stakeholders insights without engineering support.
- Developed a real-time event-processing solution that updates operational KPIs across all network generations, eliminating about 98% of manual data entry and improving reporting accuracy.
- Re-engineered a legacy XML data-processing workflow with pandas and lxml, cutting a critical processing cycle from about 6 hours to 20 minutes.
- Implemented automated data quality and validation checks that flag duplicate IDs and parameter mismatches, and rationalised 300+ unused KPIs to reduce database overhead.
- Gathered requirements from engineering and business stakeholders across 4 regional teams, translated them into analytical and automation solutions, and documented workflows for reuse.

Computer Science Teaching Assistant
University of Hertfordshire - New Administrative Capital, Egypt | Sep 2022 - Sep 2023

- Delivered a machine learning training programme to 30+ students across two cohorts, translating complex technical material into practical exercises.
- Designed and ran data science and machine learning workshops for 50+ faculty members, and held 4-6 weekly office hours for students at mixed levels.

PROJECTS

Saudi Card Spending Intelligence (SAMA data)
Python, SQL (DuckDB), Power BI | Sep 2026
Live dashboard: mellithyy.github.io/saudi-consumer-spending-intelligence

- Collected all of SAMA's point-of-sale data (312 weekly reports since May 2020, 3.8M daily figures for 60 cities) and reconciled daily to weekly totals; found a week the report undercounts by half a day.

- Built a bilingual Arabic/English Power BI dashboard and a web version with a city-by-activity spending tree to identify trends; automated checks stop the build if any total differs from the source.

Heart Disease Risk Model: 15 Years of CDC Data
Python, pandas, scikit-learn, Streamlit | Sep 2026
Live demo: heart-disease-risk-mellithyy.streamlit.app

- Built one dataset from 15 yearly CDC health surveys (2011-2025, 6.7 million adults), harmonising renamed and recoded questions; found age-adjusted heart disease fell from 7.1% to 6.3% while obesity and diabetes rose.
- Audited my 2022 model: its F1 of 0.91 came from undersampling before the train/test split (a fair test gives 0.29). Rebuilt it with time-based validation (train 2011-2023, test 2025): ROC-AUC 0.85, F1 0.42, calibrated; deployed as a Streamlit app.

Riyadh 4G Network Intelligence
Python, statsmodels, Tableau | Aug 2026
Live dashboard: mellithyy.github.io/riyadh-4g-network-intelligence

- Benchmarked three LTE operators in Riyadh on 2.7M crowdsourced measurements; removed four data quality defects (27.5% of rows) that would have inflated one operator's traffic 4.1x and flipped the ranking.
- Tested whether better coverage drives traffic with cell and hour fixed-effects regression; found no significant effect ($p = 0.57$), exposing a confounded correlation and preventing a misleading business case.
- Built 24-hour traffic forecasts per operator; SARIMA beat naive, Holt-Winters, gradient boosting and Prophet baselines (MASE 0.76, about 24% better than seasonal naive).

SQL Server Data Warehouse and Analytics (course project)
SQL Server, T-SQL | Jan 2026
GitHub: github.com/mellithyy/sql-data-warehouse-with-eda-project

- Built a SQL Server data warehouse (bronze, silver and gold layers) with T-SQL ETL, data cleansing and star-schema data models integrating ERP and CRM sources; wrote SQL analyses of customer behaviour, product performance and sales trends.

EDUCATION

MSc in Data Science
University of East London - London, United Kingdom | Sep 2024 - Present

BSc in Computers and Systems Engineering
Zagazig University - Faculty of Engineering - Zagazig, Egypt | Sep 2015 - Jul 2020
Grade: 85.16% - Excellent with Honours; ranked 3rd in department
Graduation project: Breast Cancer Diagnosis Using Machine Learning - benchmarked six classification models, reduced features from 30 to 7 while keeping 96.84% validation accuracy, and deployed the logistic regression model (96.55% accuracy) as a web application.

CERTIFICATIONS

- The Power of Statistics - Google, 2025
- Python for Data Science - IBM, 2022
- Certified Data Scientist - Epsilon AI Institute, 2022
- Designing and Implementing a Data Science Solution on Azure - Microsoft, 2022
- AI Deployment: Algorithms, Platforms and Technology - Ericsson, 2022
- Data Analysis Nanodegrees (Advanced, Professional and Challenger) - Udacity, 2021-2022

PERSONAL DETAILS

Nationality: Egyptian
Location: Al Khobar, Saudi Arabia
Iqama Status: Transferable
Driving Licence: Valid Saudi driving licence
Languages: Arabic (Native), English (Professional working proficiency)
Availability: Immediate joiner